pfi

Python for Fixed Income

COURSE 1 · PYTHON FOUNDATIONS

Course Project

WEEK 4 · DAY 5

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This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

The scenario: a new month-end extract

Month end, as of 2026-10-30

You are the analyst on the credit desk. Operations has sent the holdings extract, with a cover note.

The portfolio manager wants to know

1. What is in the book
2. Where it is concentrated
3. Whether the file can be trusted
4. How the book differs from its benchmark

A new file. Run the routine, do not remember it.

Everything in the files is invented: the issuers, the tickers, the CUSIPs, the prices, and the positions.

The cover note, word for word

```
To: Credit desk
From: Operations
Subject: Month-end holdings extract, as of 2026-10-30

The month-end extract is attached as project_holdings.csv.

Control totals for the book:
  Number of bonds:      206
  Total face:           507,250,000 dollars
  Total market value:   509,928,477.50 dollars

Market value is face times clean price plus accrued interest, divided by 100.
All positions are priced as of 2026-10-30.
Please tell us if the extract does not agree with these totals.
```

The three control totals are given to you. Test the file against them.

The note is also in the course repo, as data/project_cover_note.txt.

What you hand in

One notebook, completed, with four things in it

1. A cleaning function with asserts: question 4.2
2. An Excel workbook, two sheets: question 9.1
3. An AI-use log: question 9.2
4. A written conclusion: question 9.3

`week4/day5/course1_project.ipynb`

`course1_project_solutions.ipynb`

Every answer and a model conclusion. Open it after you have written your own conclusion.

Before you hand in

Run it all

Restart the kernel and run the whole notebook from the top.

Every check

Every cell runs, and every check prints correct.

The raw file

raw has the shape it had when it was loaded.

Every fix

A markdown cell above it gives the decision and the evidence.

The function

clean_extract has a docstring, and its three asserts pass.

No file left

The notebook deletes the workbook it writes.

The log and the page

The log is filled in. The conclusion is 250 to 350 words.

The files, and what each holds

All in the data folder of the course repo. The setup cell loads the first three.

File	What it holds	One row is	As of
`project_holdings.csv`	The month-end extract from operations	one bond the desk holds, or should be	2026-10-30
`benchmark.csv`	The benchmark of the book, 821 bonds	one bond, with its weight_pct	2026-09-30
`ratings.csv`	The rating scale, 18 rows	one rating: score, bucket, and grade	no date
`issuers.csv`	The issuer table of earlier notebooks	one issuer. The project does not load it	no date
`project_cover_note.txt`	The note, with three control totals	not a table: a note to read	2026-10-30

The benchmark file is September's. The desk has not been sent an October one, so the two files are a month apart.

Columns	Unit
`coupon, yield_pct`	percent
`oas`	basis points
`duration, spread_duration`	years
`price, accrued`	per 100 of face

Columns	Unit
`par_held, market_value`	dollars
`amount_outstanding`	dollars, the whole issue
`dts`	spread duration times OAS
`weight_pct`	percent, in the benchmark only

Nine sections, two or three sittings

Week 4, day 5 in the course map. It is more than one sitting: plan on two or three.

Sitting 1: section 1

Structure

2 questions

Rows, columns, and identifiers

Notebook 09

→

Sitting 1: section 2

Data types

2 questions

Types, a working copy, real dates

Notebook 13

→

Sitting 1: section 3

Test the extract

10 questions

The cover note and the desk's rules

Notebooks 13, 17, 08

→

Sitting 1: section 4

Fix and package

2 questions

Decisions, fixes, and a function

Notebooks 13 and 19

→

Sitting 2: section 5

Summary statistics

4 questions

Describe, skew, and counts by category

Notebooks 09, 13, 10

The first sitting ends with a table you trust and a function that rebuilds it. The other two describe the table.

Sitting 2: section 6

One column at a time

5 questions

Histograms, the IQR rule, outliers

Notebooks 11 and 13

→

Sitting 2: section 7

Two or more columns

5 questions

Correlations, scatter, pivot tables

Notebooks 11, 13, 10

→

Sitting 3: section 8

The desk's questions

10 questions

Weights, DTS, benchmark, a rate

Notebooks 10, 15, 16

→

Sitting 3: section 9

The deliverables

3 questions

Workbook, AI-use log, conclusion

Notebooks 08, 19, 01

Checkpoint after question 4.1: every line prints correct before section 5.

In green: the notebooks that taught the method. The Hints table at the end of the project notebook names the day of each.

How the work is read

For your own use: read your work against this table before you hand it in

Section	What a good answer has	Marks
1 and 2: structure and data types	the shape, the types, a working copy with real dates, and a sentence on each	8
3: testing the extract	every rule tested with a count, the failing rows shown, nothing assumed	14
4: decide, fix, and package	each decision with evidence, numbers before and after, a function with asserts	14
5: summary statistics	mean, median, and skew read together, and counts for each category	6
6: one column at a time	a finished chart for each column, far-away values examined, not deleted	12
7: two or more columns	correlations, grouped comparisons, and a pivot table, each with its count	12
8: the desk's questions	weights, concentration, weighted averages, benchmark, DTS, each with its unit	14
9: workbook, log, and conclusion	a workbook read back, an honest log, a page that never recommends	20
	Total	100

Under each question of sections 5 to 8 there is an Observations cell: write what the numbers and the chart show, with the figures and their units.

Testing the extract: the desk's rules

Test this extract against its cover note and the rules below. Fix what is an error, flag what you cannot fix, keep what is real.

Rule	What the desk expects of an extract
1	One row for each bond held. No row is a copy of another
2	A CUSIP is nine capital letters or digits, and appears once
3	The face held is positive
4	The maturity is after the as-of date
5	Market value = face x (price + accrued) / 100, within 1 dollar
6	Sector names are the names in the benchmark's list
7	No number is missing

Section 3: find

Count the rows that fail each rule. Fix nothing yet.

Section 4: decide and fix

State each decision and its evidence. Print the numbers before and after.

A clean result is a result

A rule that every row passes goes in your write-up too.

The cover note adds claims of its own: the three control totals, and that every position is priced as of the as-of date.

There is no list of what is wrong with this file. Finding out is the project.

The rules of the project

1. **Work in a copy**
   raw is never changed. Every fix is made to the working copy, book.

2. **State every decision**
   What you are changing, in how many rows, and on what evidence.

3. **Print the headline numbers**
   Before and after every fix, with headline() from the setup cell.

4. **Describe, never recommend**
   Nothing recommends, ranks, or forecasts a bond, issuer, or sector.

5. **Charts are finished charts**
   A title, a label on each axis, and the unit.

6. **The notebook runs from top to bottom**
   Restart and Run All before you hand it in.

Rule 3, as code

```
headline('Before', book)

# the fix goes here, made to book

headline('After', book)
```

What headline() prints

```
Before
   rows: ...
   face: ... dollars
   market value: ... dollars
   mean OAS: ... basis points
```

All names are invented

No name refers to a real company. Overweight and underweight name the sign of a subtraction, and nothing more.

How the checks work

A question, its empty cell, and its check cell

```
# your code here
answer_1_1 = ...

# the next cell is written for you
check('Question 1.1', answer_1_1, 'a short code')
```

check() prints	When
`Question 1.1: not attempted yet`	the answer is still ...
`Question 1.1: not yet. You have`	your answer is different
`Question 1.1: correct`	your answer matches

The right answers are not in the notebook

Each check holds a short code made from the right answer. check() makes a code from yours in the same way and compares the two. 7 and 7.0 both pass, and a decimal is rounded to the places the question states.

When a check prints not yet

1. **Read the question again**

   The unit, the rounding, and which table: raw or book.

2. **Print the headline numbers**

   Compare them with the control totals of the cover note.

3. **Still stuck after trying?**

   Some questions have a closed block. See the next slide.

4. **Made a different decision?**

   State it in a markdown cell. The checks will disagree.

A check tests the number, not the reasoning. Charts and written answers have no check.

The closed blocks

Under some questions, a closed block

> Stuck after trying? Open for the decision the checkpoint expects

It holds the decision the checkpoint expects for one kind of problem. It does not hold a count, and it does not hold an answer.

1. **Decide for yourself first**
   Leave every block closed until you have decided.
2. **Run the checkpoint**
   After question 4.1 it tests your cleaned table.
3. **Open a block only if it disagrees**
   Only the one under the question that tested that part.

Every row that fails a test is one of three kinds

An error you can fix

The files hold the evidence for the right value. Fix it, and state the evidence.

An error you cannot fix

Flag it and report it. A guess is not a fix.

A real value

Keep it, and say what shows that it is real.

A different decision can be defensible. State it, and expect the checks to disagree from there on.

Using an AI assistant, and the log

An AI assistant is allowed, used the way the course taught: four steps.

1

Read

Read the code it gives you, line by line.

2

Predict

Decide what it will print before you run it.

3

Run

Run it and look at the output.

4

Check

Compare with a number you worked out, or a control total.

An answer you cannot check is not finished.

The AI-use log, question 9.2: one row for each time you asked

What I asked	What I accepted or rejected	How I checked it
…	…	…

none used is a valid entry. Keep notes as you go.

What never goes into an AI tool, or into Colab

Client names, positions, anything from your employer, and credentials. The project can be done with an assistant only because every file in it is synthetic. Your firm's policy comes first.

The written conclusion

Question 9.3: the page for the portfolio manager, in 250 to 350 words

Part 1

Five numbers

The five figures a reader needs first, each with its unit.

Part 2

Concentrations

Where the market value and the DTS sit: by sector, grade, and issuer.

Part 3

Data quality

Each problem, in how many rows, and what was done. What was found clean.

Part 4

The benchmark

Differences in weight by sector and rating bucket, and the two file dates.

Part 5

Open items

What the operations team needs to confirm or send.

Describe. Do not recommend, rank, or forecast.

Overweight and underweight mean a positive or a negative difference in weight, and nothing else.

There is no check cell for this question.

A good conclusion

Can be read without the notebook

Has a number in almost every sentence

Gives the unit of each number

Holds nothing that a cell above does not support

What you can do now

- Write Python: variables, loops, functions, classes, and reading an error
- Hold a portfolio in pandas: filter it, group it, join it, and summarize it
- Draw finished charts: a title, labelled axes, and units
- Test a new file before you trust it, and package the fixes as a function
- Write up what the data shows, without a recommendation in it

Course 2 Bond Math: pricing, yield, duration, and convexity, built by hand and checked against Excel